

**AMENDMENT NO. 2 DECEMBER 2002
TO
IS 11487 : 1985 SPECIFICATION FOR SODIUM
LAURYL ETHER SULPHATE FOR
COSMETIC INDUSTRY**

(Page 3, clause 2.1.1, line 2) — Substitute 'liquid' for 'solution'.

[Page 4, Table 1, Sl No. (xii) (see also Amendment No. 1)] — Insert the following after Sl No. (xii):

(1)	(2)	(3)	(4)
xiii)	1,4 Dioxane, on 100% basis, ppm, Max	100	A-14

(Page 6, clause A-2.1.2.1, line 6) — Substitute 'conical/iodine flask' for 'beaker'.

(Amendment No. 1, clause A-13.4) — Insert the following new clause after A-13.4:

A-14 DETERMINATION OF 1, 4 DIOXANE BY GAS CHROMATOGRAPHY METHOD UTILISING HEAD SPACE TECHNIQUE

A-14.1 Outline of the Method — Gas chromatographic method for the estimation of the presence of 1, 4 dioxane in sodium lauryl ether sulphate is given below both for packed column and capillary column. In case of dispute, Gas Chromatographic method using capillary column shall be considered as referee method.

A-14.2 Principle

The head space technique takes advantage of the volatility of 1,4 dioxane over the non-volatile ionic constituents of ethoxylated alcohol sulfate. Only the vapours above the liquid sample are injected onto the chromatographic column, preventing contamination of the column as compared to the direct injection techniques. Reproducibility is ensured by the use of an internal standard added to the sample with a retention time similar to 1,4 dioxane. The relative volatility of 1,4 dioxane is increased by adding N,N-dimethyl formamide to the sample matrix enhancing the sensitivity of the technique.

Amend No. 2 to 11487 : 1985

A-14.3 Gas Chromatographic Method Using Packed Column

A-14.3.1 Apparatus

- a) Any gas chromatograph equipped with head space analyser, flame ionization detector (FID), and data handling package including CRT screen and printer.

The GC operating conditions are given below for guidance only:

<i>Column</i>	Packed		
<i>Material</i>	Stainless steel		
<i>Length</i>	6 feet		
<i>I.D</i>	1/8 inch		
<i>Stationery Phase</i>	Tenax >60/80 mesh		
<i>Carrier gas</i>	Helium, 30 ml/min		
<i>Temperatures</i>	Temp (°C)	Time (min)	Rate (°C/min)
Headspace Sampler	70	30	
First Isothermal	Oven 170	6	
Oven Ramp Rate			30
Second Isothermal	Oven 240	4	
Injector	290		
Flame Ionization Detector	290		

Others

FID Sensitivity — High

Timed Events for Headspace Sampler

Amend No. 2 to 11487 : 1985

Time (min)	Event	Position
-1.01	Hold	
-0.01	Relay 0	on
0.00	Relay 1	on
0.10	Relay 1	off
0.11	Relay 0	off

Data Acquisition

Start time	2.50 minutes
End Time	6.00 minutes
Width	-4
Detector Area Sens	25 to 500
Detector Base Sens	10 to 15
Calculation Type	Internal standard
1,4-Dioxane Retention Time (min)	3.1
Toluene Retention Time (min)	4.1

The calibration is accomplished by substituting 30% active ethoxylated alcohol sulfate with 30 percent active sodium lauryl sulfate and in place of the internal standard solution, a calibration solution is used. The details of calibration solutions are as follows:

Calibration solution — 50 mg/litre 1,4-dioxane + 50 mg/litre toluene in N,N-dimethylformamide.

Internal Standard — 50 mg/litre toluene in N,N-dimethylformamide

The standard analysis procedure is as follows:

b) Analytical balance — with accuracy of 0.0001 g.

Amend No. 2 to IS 11487 : 1985

A-14.3.2 Reagents

A-14.3.2.1 1,4-dioxane — 99.9% purity.

A-14.3.2.2 Toluene — HPLC Grade

A-14.3.2.3 N,N-dimethyl formamide — 99.9% purity.

A-14.3.2.4 Sodium lauryl sulfate, 30% active — commercial material.

A-14.3.3 Procedure

0.5 g sample of 30 percent active ethoxylated alcohol sulfate is weighed into a vial with an accuracy of 0.1 mg. To the sample is added 1 ml of internal standard solution. The sample is sealed with a teflon faced septum using the crimping device. The sample vial is inserted into the heating block, equilibrated there for 30 minutes at 70°C. The chromatographic run is initiated manually by pushing a fixed injection needle through the septum of the vial, the valve relays direct the sample and helium flow. The result is that a portion of the vapour phase from the vial is injected onto the GC column. As the components of the sample vapour are resolved and exhausted to the FID the analog output is converted to digital output and collected by the data handling package. The toluene and 1,4 dioxane peaks are identified by retention times and the quantity of 1,4-dioxane is calculated from the relative peak areas of toluene and 1,4-dioxane, the result is output to the screen and printer along with a copy of the chromatogram. If any significant amount of a solvent such as ethanol is present in the ethoxylated alcohol sulfate sample then the calibration samples must include a like amount of the solvent. The chemical matrix of the sample effects the relative volatility of 1,4-dioxane and toluene and consequently the relative peak areas.

A-14.3.4 Calculations

Calibration factor F

$$F = \frac{\text{mg (1,4-dioxane)} \times \text{peak area (toluene)}}{\text{mg (toluene)} \times \text{peak area (1,4-dioxane)}}$$

1,4-dioxane in ppm (mg/kg)

$$\text{1,4-dioxane} = \frac{\text{mg (toluene)} \times \text{peak area (1,4-dioxane)} \times F \times 10^6}{\text{mg (sample weight)} \times \text{peak area (toluene)}}$$

Amend No. 2 to 11487 : 1985

A-14.4 Gas Chromatographic Method Using Capillary Column

A-14.4.1 Outline of the Method

The method describes the determination of 1,4-dioxane in alcohol ethoxysulphates up to 1 000 mg/kg.

A weighed amount of sample is analyzed after stabilization in a heated compartment using a Headspace sampler, by means of gas chromatography over a fused silica capillary column using a sample splitter and flame ionization detection. This method describes the determination of 1,4-dioxane in alcohol ethoxysulphates up to 1000 mg/kg.

A-14.4.2 Apparatus

A-14.4.2.1 Any Gas Chromatograph provided with a flame ionization detector, a sample splitter, temperature-programming facilities and a Headspace sampler.

Operating Instructions for Gas Chromatograph

Column

Type	Capillary
Material	Fused Silica
Length, I.D.	60 m × 0.32 mm
Stationary Phase	Methyl silicone fluid
Film thickness	1 micron

Detector

Type	Flame Ionization
Sample Splitter	Split ratio 1:30

Temperatures

Column Oven	Initial temperature: 80°C hold for 7 minutes Final temperature: 250°C hold for 20 minutes Programming rate: 15°C/min
Injector	250°C

Amend No. 2 to IS 11487 : 1985

Head space sampler	Aux. temperature : 80°C
Detector	300°C
FID sensitivity	High
FID zero	On
Carrier Gas	Helium 1 ml/min
Pressure	120 kPa

Time in Headspace : 15 minutes

Sampler

Time events

Time : 1.01 min. Event : hold

Time : 1.00 min. Event : relay 0 : on

Time : 0.00 min. Event : relay 1 : on

Time : 0.10 min. Event : relay 1 : off

Time : 0.11 min. Event : relay 0 : off

A-14.4.2.2 Injection vials 10 ml, with matching serum caps for safe closing.

A-14.4.3 *Reagents and Materials*

A-14.4.3.1 *1,4-Dioxane, A.R.*

Caution: 1,4-Dioxane presents a health hazard if incorrectly handled. Prevent contact with the skin. Avoid inhalation.

A-14.4.4 *Procedure*

A-14.4.4.1 Into two separate vials, accurately weigh 2 g of homogeneous sample, to the nearest 10 mg. Introduce an amount 1,4-dioxane to one of the vials to obtain an accurately known concentration of added 1,4-dioxane of approx. 100 mg/kg, and mix thoroughly. Seal both vials immediately in such way that no leakage can occur.

NOTE — Dilute weighed samples with water, if necessary and apply correction factor in calculations.

A-14.4.4.2 Place the vials subsequently into the head space sampler and warm for 15 minutes at 80°C and inject into the gas chromatograph according to the operating instructions given.

Amend No. 2 to IS 11487 : 1985

A-14.4.4.3 Determine the active matter content of the sample in % (m/m) as per procedure given in IS 4956 : 2002 'Synthetic detergents for industrial purposes (second revision)'.

A-14.4.5 *Interpretation of the Chromatogram*

The peak for 1,4 chromatogram appears at about 3 to 3.5. A typical chromatogram under the conditions given above is shown in Fig. 1.

A-14.4.6 *Calculation*

Calculate the 1,4-dioxane content of the sample by means of the following equation:

$$\text{1,4-dioxane, mg/kg} = \frac{A}{B-A} \times \frac{100}{C} \times K$$

where

A = peak area of 1,4-dioxane in sample without addition, arbitrary units;

B = peak area of 1,4-dioxane in sample after addition, arbitrary units;

C = active matter content of sample, percent (m/m);

D = concentration of 1,4-dioxane added to sample, mg/kg; and

K = sample dilution factor (if applicable).

Amend No. 2 to IS 11487 : 1985

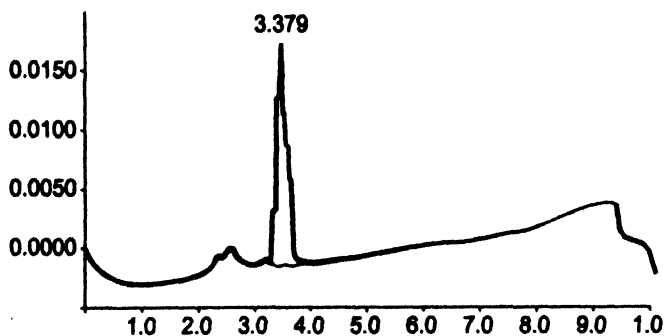
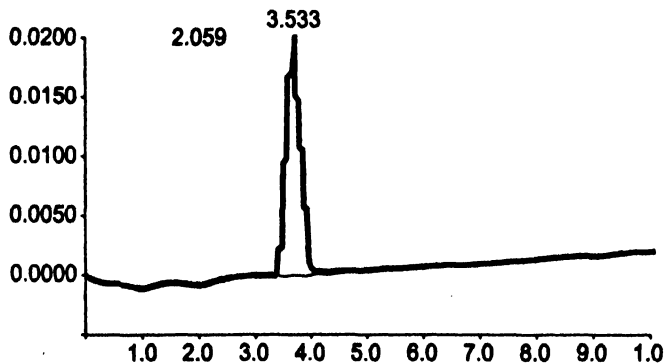


FIG. 1 A TYPICAL CHROMATOGRAM FOR 1, 4 DIOXANE

(PCD 19)